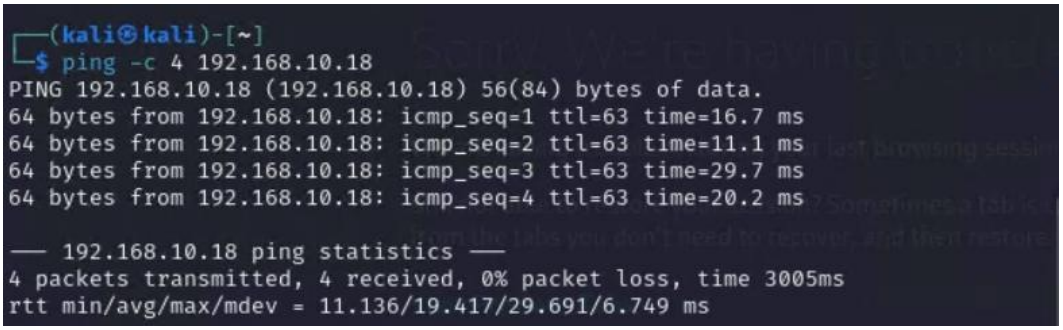
Lab Report

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Task 1: Blocking ICMP Traffic

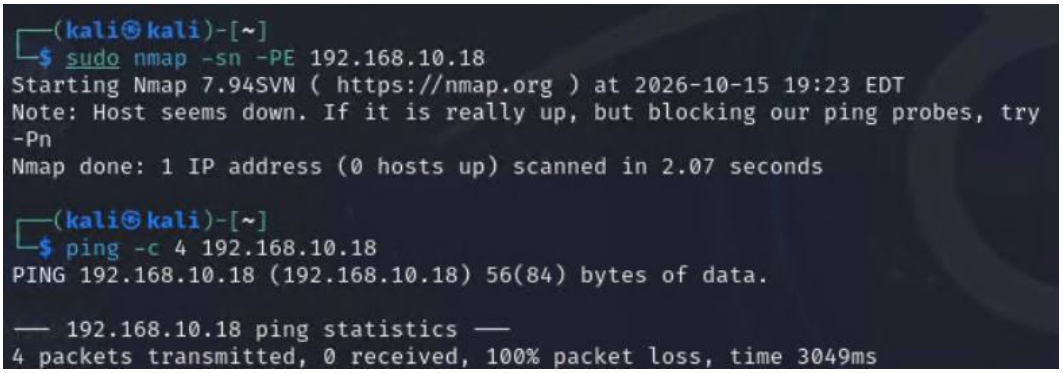
First I powered on the External Kali, pfSense and Ubuntu VMs. When I tried to ping Ubuntu at 192.168.10.10 it kept saying the host was down. After some troubleshooting I ran ip a on Ubuntu and found out it actually got 192.168.10.18 from DHCP instead of .10, so I used .18 for the rest of the task. Before adding any rule, Kali could ping Ubuntu with no packet loss. The ttl=63 shows the traffic went through pfSense.



Next I logged into the pfSense web interface from Ubuntu and went to Firewall > Rules > WAN. I added a rule for IPv4 ICMP with source 192.168.217.3 (Kali) and destination 192.168.10.18 (Ubuntu). The first time I saved it with the action set to Pass by mistake (the green check in the screenshot), so the ping still worked. I edited the rule, changed the action to Block and applied the changes.



After the Block rule was applied, the Nmap ping scan said the host seems down and the ping had 100% packet loss, so the rule worked.

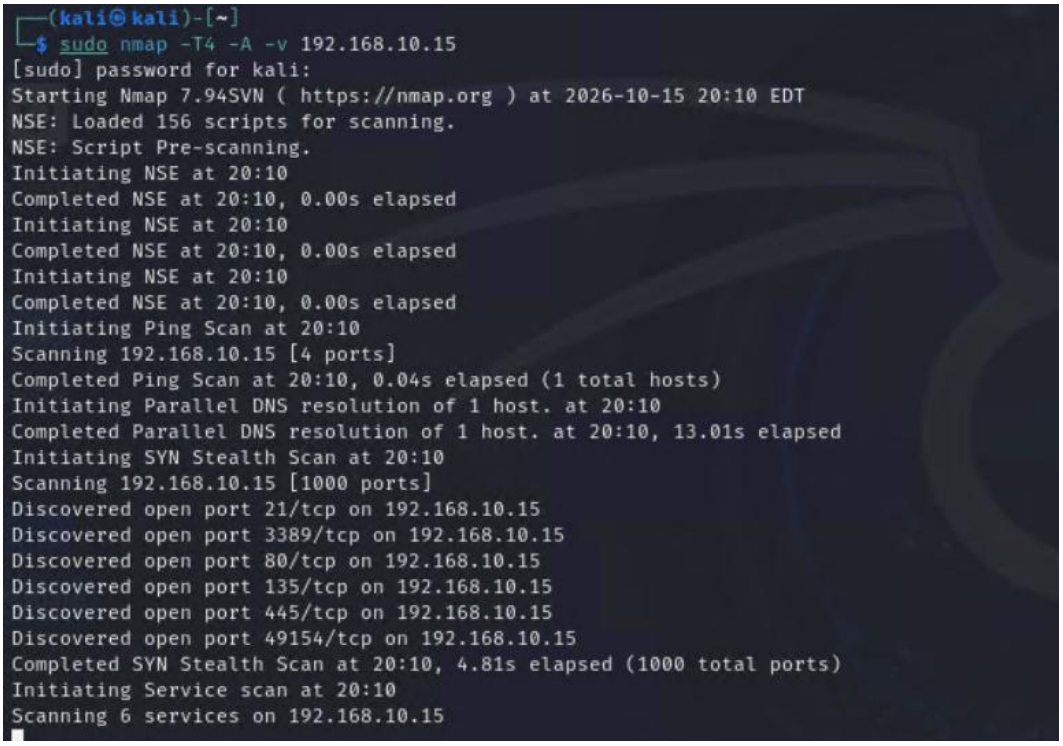


Task 2: Blocking FTP Traffic

For this task I also powered on the Windows Server 2008 VM. It was not at 192.168.10.11 like the instructions said, so I scanned the subnet for port 21 with nmap -Pn -p 21 --open 192.168.10.0/24 and found it at 192.168.10.15. Also the instructions say to block FTP to the Ubuntu VM, but the FTP server is the Windows Server so I used that as the destination. Before the rule, FTP connected to the server and showed the Microsoft FTP Service banner.



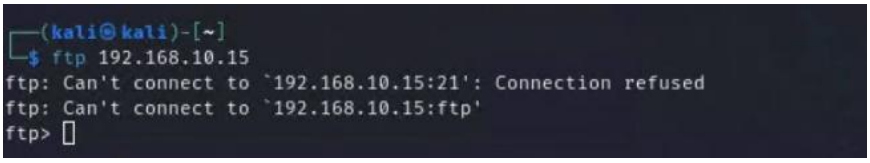
The intense scan (nmap -T4 -A -v) also showed port 21/tcp open along with some other ports like 80, 135, 445 and 3389.



Then I added a rule on the WAN interface to reject TCP traffic from 192.168.217.3 to 192.168.10.15 on destination port 21 (FTP). I used Reject instead of Block so the connection gets refused right away, which is why the port shows as closed instead of filtered.



After applying the rule, the FTP connection from Kali was refused, so FTP traffic from the WAN was blocked.



Reflection

This lab showed me how much control a firewall gives you over specific kinds of traffic. ICMP seems harmless since it's mostly used for ping, but attackers use ping sweeps to find out which hosts are alive before they scan them further. Blocking ICMP from the outside made the Ubuntu VM look like it was offline to Nmap, which makes recon harder. ICMP can also be used for things like ping floods and tunneling data out of a network. You don't want to block every type of ICMP everywhere though, because some of it is needed for normal network functions.

FTP is a bigger risk because it sends usernames, passwords and files in plain text, so anyone sniffing the network can read them. The server in this lab even had anonymous login enabled, which means anyone from the WAN could have connected. Blocking port 21 removes that exposure, and if file transfer is needed it should be done with something encrypted like SFTP or FTPS.

I also learned that the small details matter. My first ICMP rule was set to Pass instead of Block and it did nothing, and the rule order matters since pfSense uses the first rule that matches. It's important to always test the rule afterwards to make sure it actually works. Overall, only allowing the traffic that's really needed and blocking the rest is a simple way to shrink the attack surface of a network.